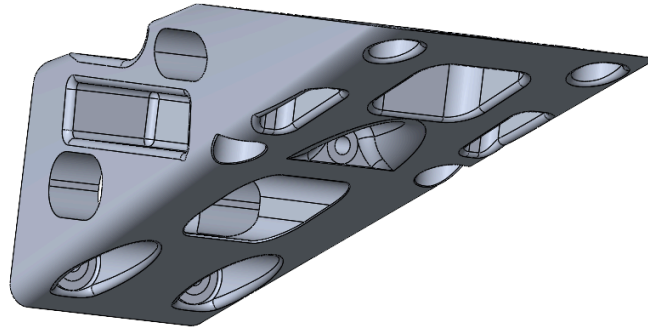


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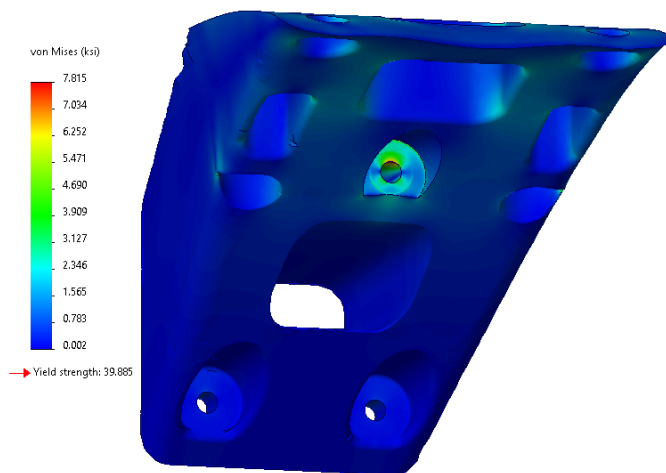
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CAE Final Project — Optimized Structural Connector

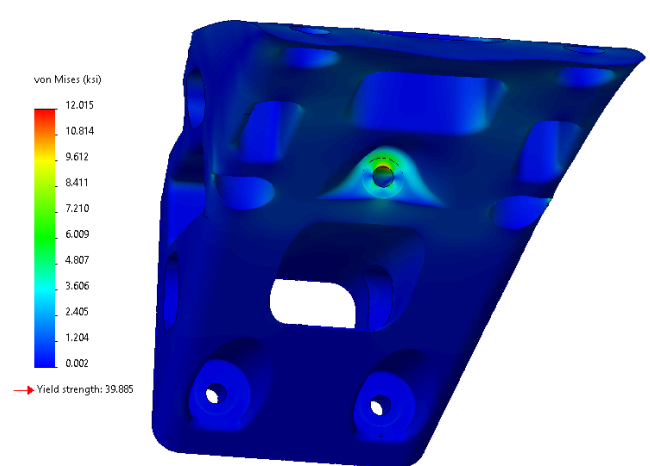
Fully self-modeled in SolidWorks. Brief: bridge two mounting profiles in 6061-T6 aluminum under a 2,000 lbf distributed load, maximizing FOS-to-weight. Five design iterations, each checked with a converged h-adaptive static study. A topology study guided where material came out.



Final design (Iteration 5 configuration). Webbed connector with pockets cut where the topology study showed low stress. Fully parametric, with a configuration saved for each iteration.



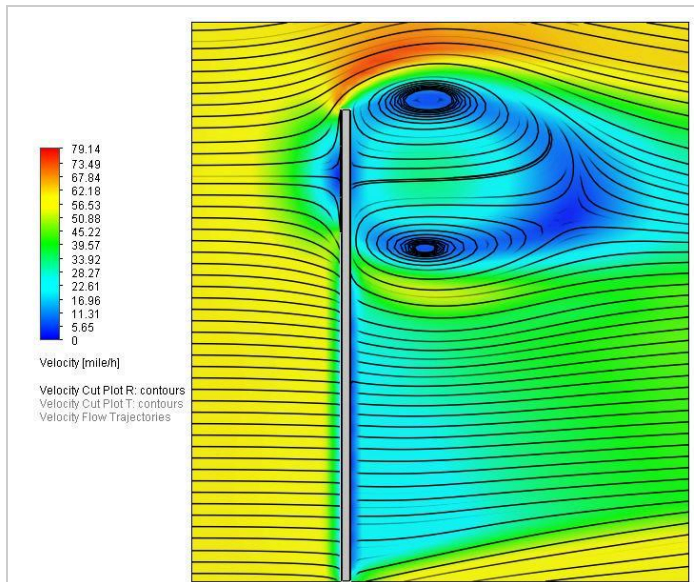
Static study, earlier iteration. von Mises stress, max 7.8 ksi against 39.9 ksi yield (6061-T6), converged h-adaptive mesh.



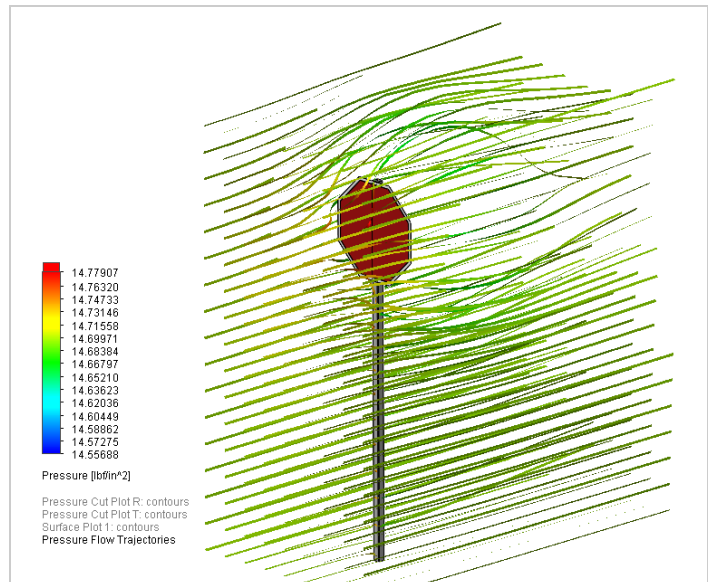
Static study, later iteration. After more material came out, peak stress rose to 12.0 ksi. Still well under yield, with less mass.

Simulation coursework

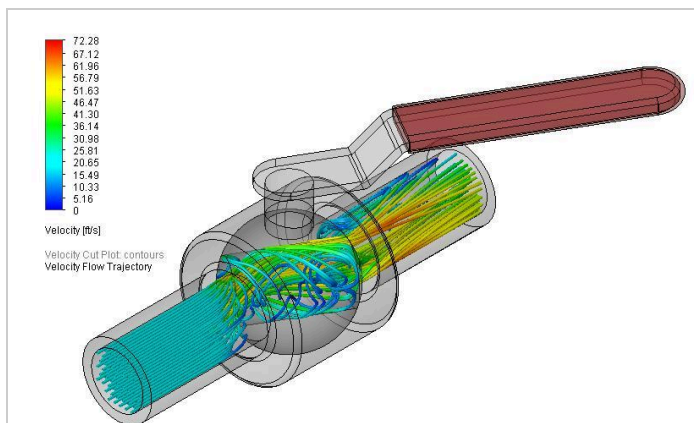
SolidWorks Flow Simulation studies from CAE coursework. Geometry provided by the instructor; flow study setup, meshing, goals, and analysis are my work. Figures below are from my submitted reports.



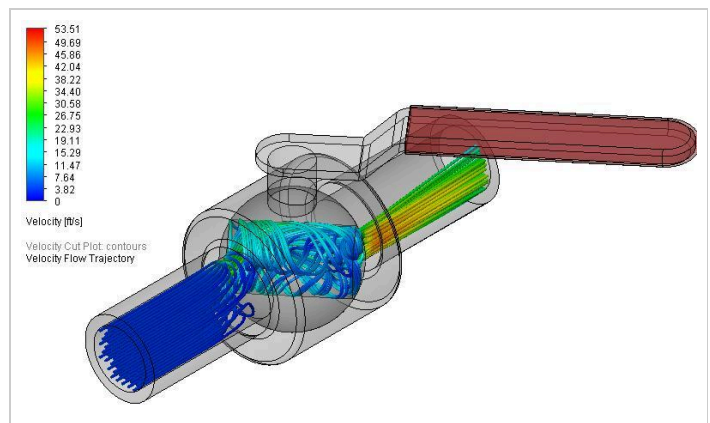
External flow over a stop sign. Velocity cut plot with flow trajectories showing wake recirculation and vortex shedding.



Same study, 3D pressure flow trajectories. Stagnation pressure on the sign face; the wake forms downstream.



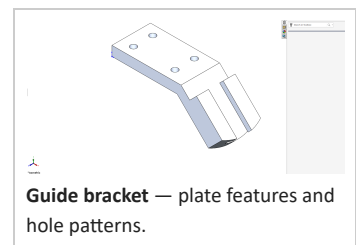
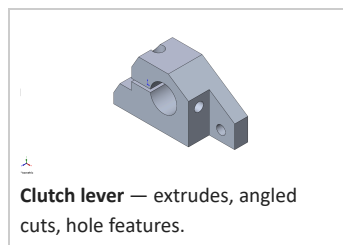
Internal flow through a ball valve, fully open. Velocity flow trajectories, peak 72 ft/s through the bore.



Same valve, throttled. Flow recirculates in the ball cavity and the jet speeds up through the smaller opening.

Parts modeling

Self-modeled parts from my introductory SolidWorks coursework, shown with feature history in SolidWorks.



Additional design work: reverse-engineered a Philips water flosser with full drawings and a technical report; designed and built a model car with passive auto-steering including a 3D-printed SolidWorks spoiler.